Group 11

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STA4365

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Evaluating Streaming Platform Longevity Through Predictive Modeling

Introduction/Background:

Streaming platforms have become one of the primary ways consumers access entertainment, but the rapid growth of subscription options has also made the market more crowded, expensive, and competitive. Research from the University of Florida found that 91% of streaming subscribers maintain multiple subscriptions, averaging 4.88 services, and that Netflix, Amazon Prime Video, and Hulu serve as the most common core platforms in these subscription portfolios. This suggests that streaming services no longer compete only to be the single "best" option, but instead to secure a strong and lasting role within consumers' broader streaming repertoires. In this project, we analyze title-level measures such as popularity, ratings, and user engagement to evaluate which major streaming platforms appear to have the strongest long-term potential. By comparing these indicators and applying predictive models, we aim to identify which services appear best positioned for sustained strength in the streaming entertainment market.

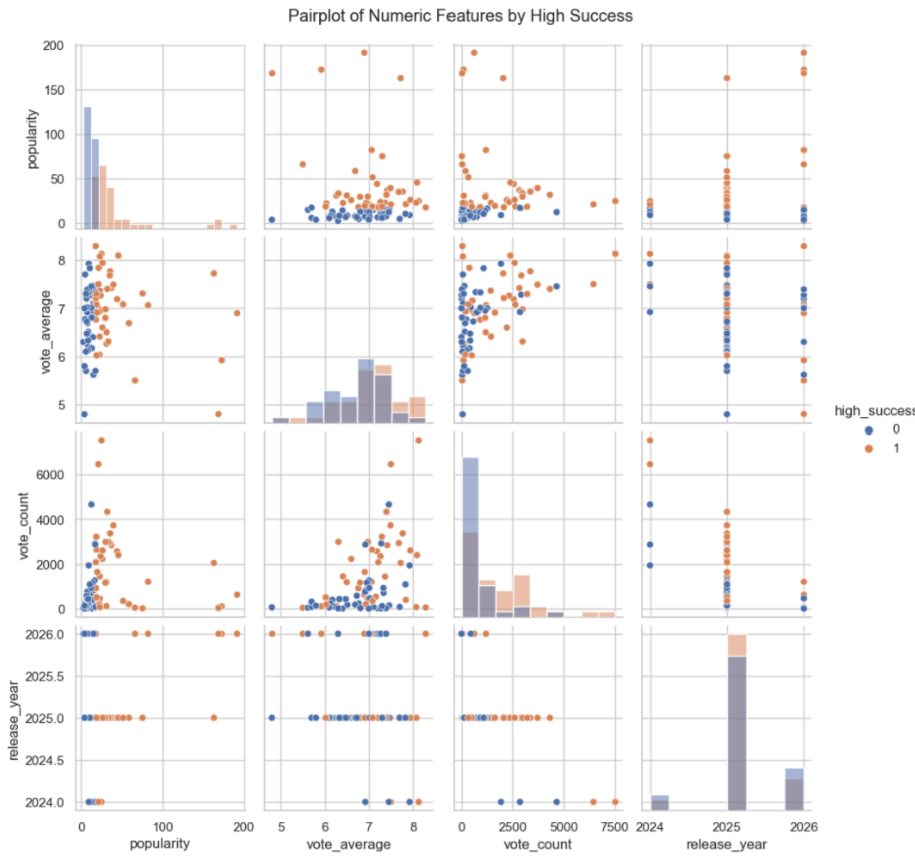
<u>Exploratory Data Analysis</u>:

This dataset was found on Kaggle. The following is the first few lines of the dataset.

	movie_id	title	release_date	popularity	vote_average	vote_count	streaming_platforms	on_netflix	on_hulu	on_prime
0	1317672	Love Me Love Me	2026-02-12	172.5990	5.922	103	Amazon Prime Video, Amazon Prime Video with Ads	0	0	1
1	1242898	Predator: Badlands	2025-11-05	163.0384	7.723	2041	Disney Plus, Hulu	0	1	0
2	1243460	Joe's College Road Trip	2026-02-12	66.1207	5.500	39	Netflix, Netflix Standard with Ads	1	0	0
3	1426964	State of Fear	2026-02-10	168.5810	4.806	31	Netflix, Netflix Standard with Ads	1	0	0
4	1208348	Rental Family	2025-11-20	21.4637	7.839	388	Hulu	0	1	0

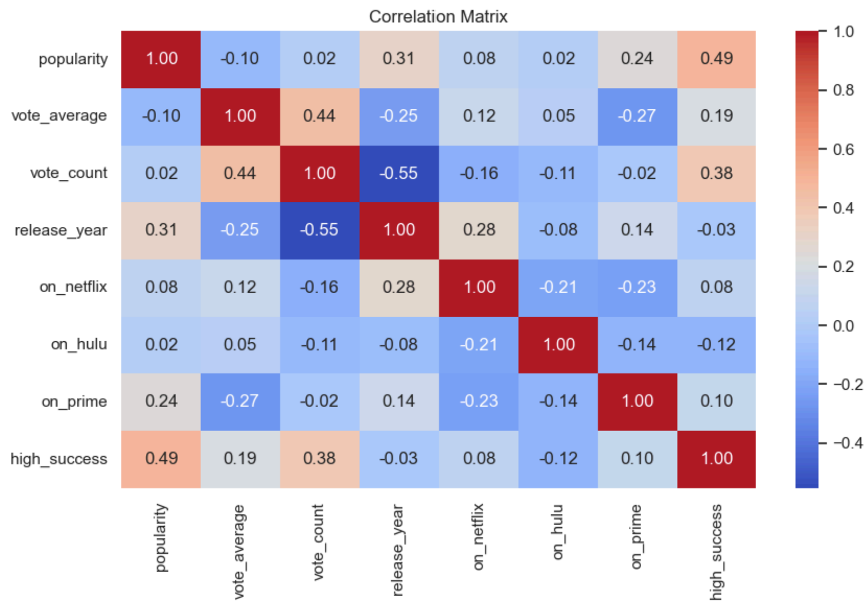
The dataset is a structured movie-level dataset capturing information on titles distributed across the 3 major streaming platforms. Each observation represents an individual movie and includes descriptive, performance, and platform-related variables. The descriptive variables include movie_id, title, and release_date, while the performance related variables include popularity, vote_average, and vote_count, which serve as indicators of audience attention, reception, and engagement. The dataset also includes a streaming_platforms field and binary platform indicators (on_netflix, on_hulu, and on_prime) that identify where each title is available. Together, these variables make the dataset useful for both comparative analysis and predictive modeling, since they allow us to examine how title performance differs across streaming platforms and to estimate which platforms appear strongest based on the success patterns of their content.

Pairplot of Numeric Features by High Success



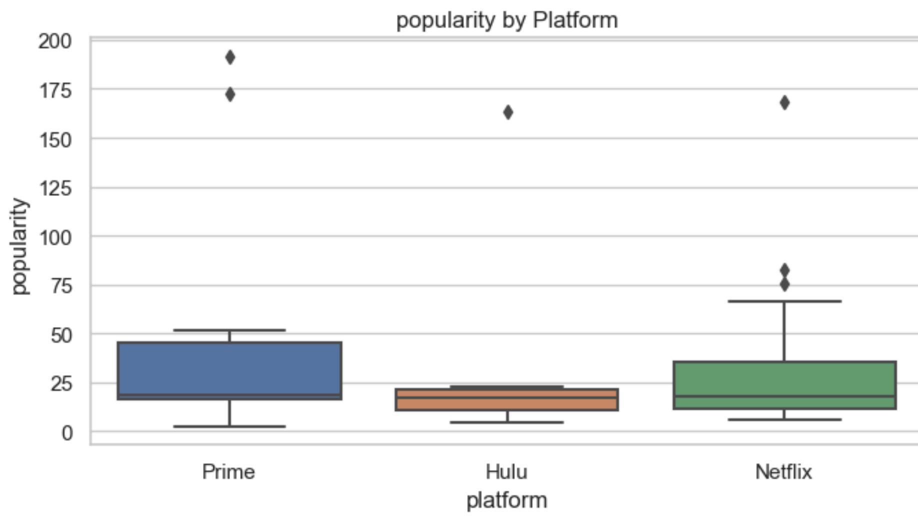
Within the pairplot of numeric features by success, there is no single strong linear relationship across all variables, but clear patterns still appear. The strongest separation between high-success and low-success titles is in popularity and vote count, where high-success titles tend to have higher values. This suggests that audience attention and engagement are the strongest indicators of title success. Vote average shows some separation, but there is still substantial overlap between the two groups, meaning ratings are helpful but less influential. Release year shows little clear relationship with success, since both groups appear across the same years. The distributions also show that popularity and vote count are strongly right-skewed, indicating outliers and suggesting that log transformations may improve modeling. Overall, the pairplot suggests that popularity and vote count are the most important predictors of success, while vote average has moderate value and release year contributes less.

Heatmap/Correlation Matrix

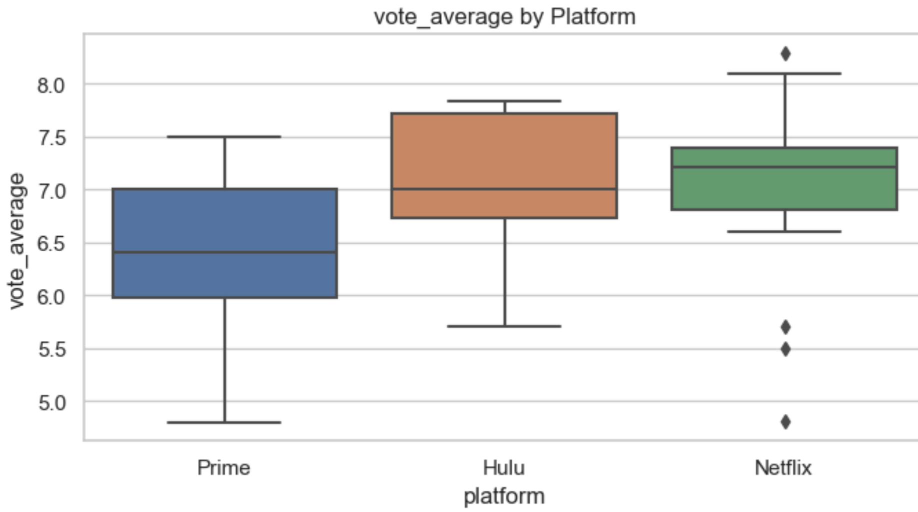


The correlation/heatmap matrix shows that most variables have only weak to moderate relationships, suggesting that no single feature fully explains title success on its own. The strongest positive correlation with high_success is popularity at 0.49, followed by vote_count at 0.38, which indicates that audience attention and engagement are the most important predictors of success in this dataset. Vote_average has a weaker positive correlation with high success at 0.19, meaning ratings provide some useful information but are less influential than popularity or vote count. In contrast, release_year has almost no relationship with high success at -0.03, showing that recency alone does not meaningfully predict success. Among the predictors, vote_average and vote_count have a moderate positive correlation of 0.44, while vote_count and release_year have a notable negative correlation of -0.55, suggesting that older titles may have had more time to accumulate votes. The platform variables show only weak correlations with high success, with on_prime at 0.10, on_netflix at 0.08, and on_hulu at -0.12, indicating that platform membership alone is not a strong direct predictor. Overall, the correlation matrix supports the earlier findings that popularity and vote count are the strongest indicators of title success, while ratings have moderate value and platform variables contribute less directly.

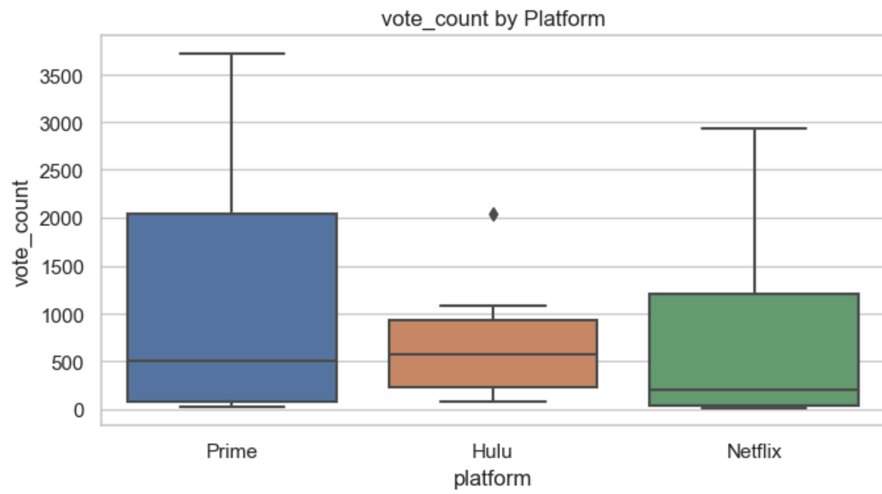
<u>Box Plots</u>



The boxplot of popularity by platform shows that Prime and Netflix generally have higher popularity levels than Hulu, while Hulu's titles appear more concentrated at lower popularity values. Netflix shows the widest spread overall, indicating greater variation in title popularity, and also includes several high-popularity outliers, suggesting that it carries a few especially popular titles alongside many moderate-performing ones. Prime also shows a relatively wide spread and contains some of the highest outliers in the dataset, which suggests strong upside in title popularity. In contrast, Hulu has a narrower distribution and lower central tendency, showing more consistently modest popularity levels with fewer highly popular titles. Overall, the graph suggests that Prime and Netflix tend to host stronger-performing titles in terms of popularity, while Hulu lags behind in this measure.

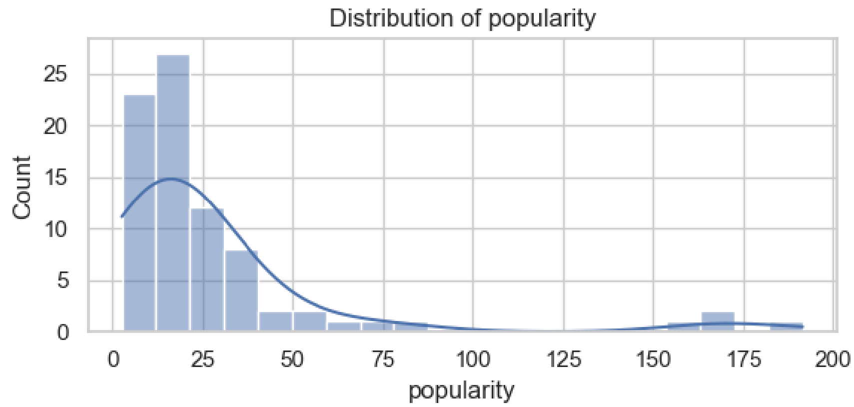


The boxplot of vote average by platform shows that Netflix and Hulu generally have higher audience ratings than Prime, while Prime's titles tend to have a lower central rating overall. Netflix has the highest median vote average, suggesting that its titles are rated slightly better on average, although it also includes several low-rating outliers, indicating a few poorly rated titles within the platform. Hulu also shows a relatively strong median rating with a somewhat wider upper range, suggesting that many of its titles are consistently well rated. In contrast, Prime has the lowest median and a broader lower range, indicating more variation toward lower-rated content. Overall, the graph suggests that Netflix and Hulu perform better than Prime in terms of average audience ratings, with Netflix showing the strongest overall rating profile.

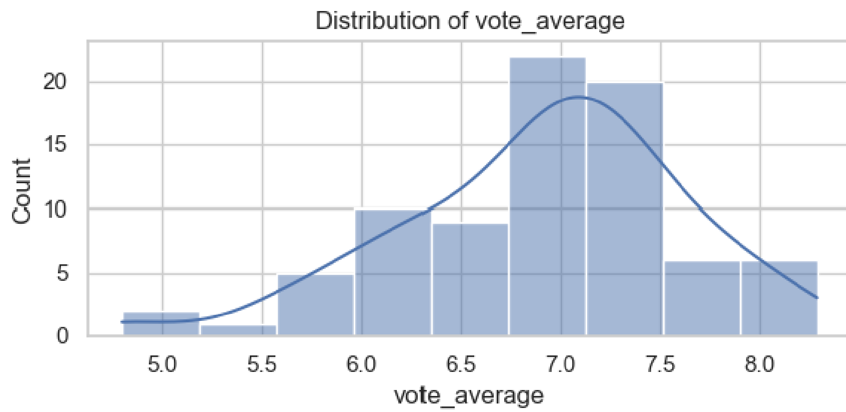


The boxplot of vote count by platform shows clear differences in audience engagement across the three streaming services. Prime has the highest overall spread and the highest upper range, indicating that while many of its titles receive relatively few votes, it also includes several titles with very large vote counts, suggesting strong engagement for some of its content. Netflix also shows a wide distribution, but with a lower median vote count than Hulu, meaning its engagement levels vary greatly across titles. Hulu, in contrast, has the highest median vote count, which suggests that its titles tend to receive more consistently moderate audience engagement, even though it has a narrower overall range and fewer extremely high-vote titles than Prime. Overall, the graph suggests that Prime and Netflix have greater variability and some highly engaged titles, while Hulu shows more consistent vote count performance across its catalog.

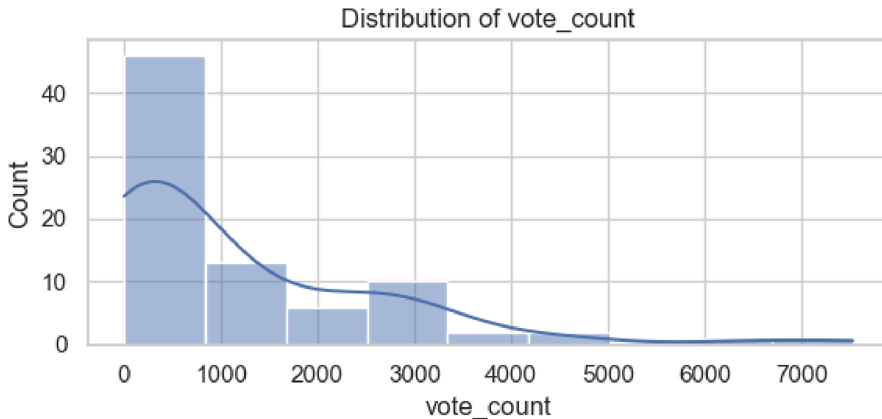
Histograms



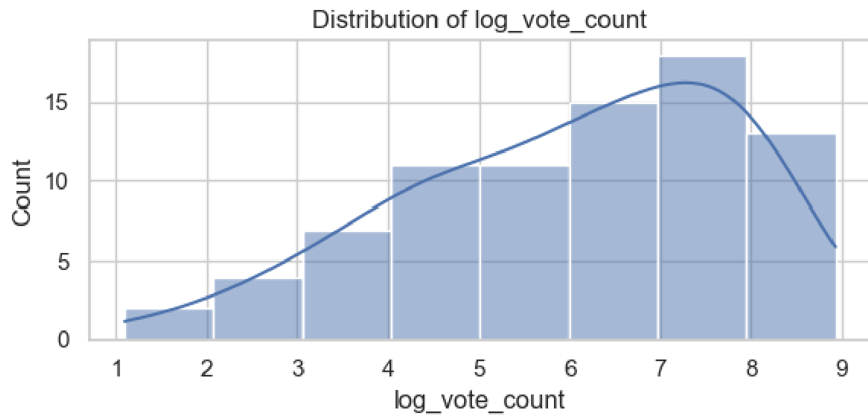
The histogram of popularity distribution shows that the variable is strongly right-skewed, with most titles concentrated at relatively low popularity values and only a small number of titles reaching very high levels. The highest concentration appears roughly in the lower range of the distribution, indicating that most movies in the dataset attract modest popularity rather than extreme attention. At the same time, a long right tail and a few very large values suggest the presence of outliers, meaning a small set of titles are substantially more popular than the rest. This pattern implies that popularity is not normally distributed and may benefit from a log transformation before modeling to reduce the influence of extreme observations. Overall, the graph indicates that while most titles have moderate or low popularity, a few standout titles account for much of the upper end of the distribution.



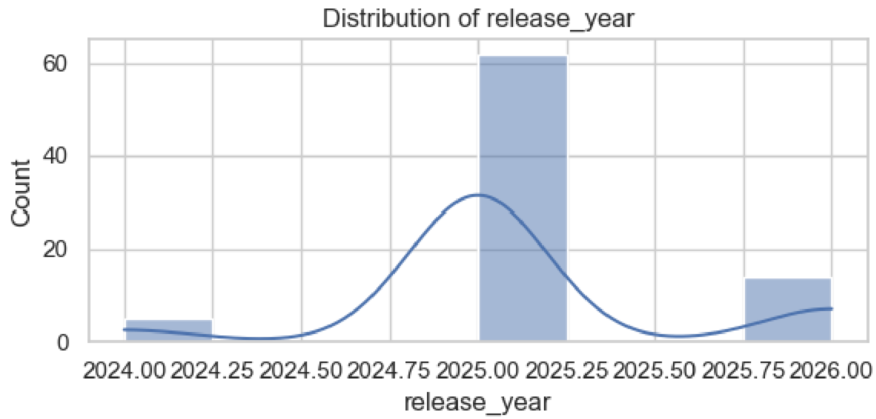
The histogram of vote_average shows a more concentrated distribution centered around 7.0, meaning that most titles in the dataset receive moderate to moderately high audience ratings. Compared with popularity and vote count, this variable appears much less skewed and more evenly distributed, although there is still some spread on both the lower and higher ends. A small number of titles fall below 5.5, while another smaller group reaches values above 8.0, suggesting limited outliers but no extreme imbalance. Overall, the graph suggests that vote average is relatively stable across titles, with most movies receiving ratings clustered in a narrow middle range, which may make it a useful but less highly differentiating predictor than more variable measures like popularity or vote count.



The histogram of vote_count shows a strongly right-skewed distribution, with most titles receiving relatively low numbers of votes and only a small number of titles attracting very high levels of audience engagement. The highest concentration is in the lower end of the distribution, indicating that the majority of movies in the dataset have modest vote counts. At the same time, the long right tail and a few very large values suggest the presence of substantial outliers, meaning that a limited number of titles receive disproportionately more audience interaction than the rest. This pattern indicates that vote count is not normally distributed and may benefit from a log transformation before modeling in order to reduce the impact of extreme values. Overall, the graph suggests that while most titles receive low to moderate engagement, a few highly voted titles drive much of the variation in the dataset.

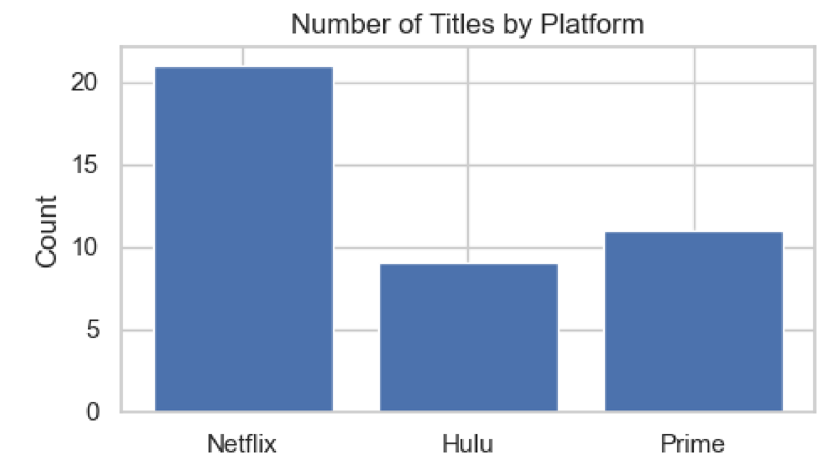


The histogram of log_vote_count shows that the logarithmic transformation substantially reduces the strong right skew that was in the original vote count distribution. After transformation, the values are spread more evenly across the range, with a smoother and more balanced shape and far less dominance from extremely large observations. This shows that the log transformation helps compress the influence of very highly voted titles while preserving differences among lower and moderate vote counts. As a result, the transformed variable is more suitable for statistical analysis and predictive modeling because it reduces the impact of outliers and creates a distribution that is easier for many models to handle. Overall, the graph suggests that log_vote_count provides a more stable and interpretable version of audience engagement than the raw vote count variable.



The histogram of release_year shows that the dataset is heavily concentrated in 2025, with only a small number of titles coming from 2024 and 2026. This means that most observations are drawn from a narrow time window, meaning the dataset primarily reflects a recent snapshot of streaming content rather than a broad historical range. Because the vast majority of titles fall within the same year, release_year has limited variation and is less likely to serve as a strong predictor on its own. Overall, the graph suggests that recency is relatively uniform across the dataset, which helps explain why release year showed only weak relationships with success in the earlier analyses.

Distribution of Data



The bar chart of the number of titles by platform shows that Netflix has the largest representation in the dataset, followed by Prime, while Hulu has the fewest titles. This indicates that the dataset is not evenly balanced across platforms, with Netflix contributing a much larger share of observations than the other services. As a result, Netflix may have a stronger influence on some descriptive summaries and model patterns simply because it is represented more often. Overall, the graph suggests that Netflix has the broadest title presence in this sample, while Hulu's smaller count means its platform-level results should be interpreted with slightly more caution due to the lower number of observations.

EDA Findings

Based on the exploratory data analysis, Prime appears to show the strongest overall long-term investment signal, with Netflix close behind and Hulu ranking weakest in this dataset. The EDA graphs showed that Prime and Netflix generally had stronger popularity distributions than Hulu, and both platforms included more high-popularity titles and stronger engagement patterns. Netflix stood out by having the highest audience ratings overall, while Prime showed particularly strong performance in popularity and broad variation in vote count, this suggests the presence of several highly engaging titles. The pairplot and correlation matrix also showed that popularity and vote count were the variables most strongly associated with high-success titles, while platform membership alone had only weak direct relationships with success. This suggests that the most promising platforms are those carrying titles with stronger audience attention and engagement, rather than those identified by name. Overall, the EDA supports the conclusion that Prime appears to be the strongest content-based investment candidate, Netflix remains a close second, and Hulu appears less competitive in this sample.

<u>Data Processing</u>:

```
Missing values:
movie_id              0
title                 0
release_date          0
popularity            0
vote_average          0
vote_count            0
streaming_platforms   0
on_netflix            0
on_hulu               0
on_prime              0
release_year          0
log_vote_count        0
log_popularity        0
high_success          0
dtype: int64

Class balance for high_success:
high_success
1    41
0    40
Name: count, dtype: int64
```

The dataset was first loaded into Python and inspected to understand its structure, variables, and overall shape. Each row represented one movie title, and the dataset included descriptive variables such as movie_id, title, and release_date, along with performance-related variables such as popularity, vote_average, and vote_count. It also included platform indicators for Netflix, Hulu, and Prime.

Before modeling, we checked the dataset for missing values. The missing value output showed that there were no missing values in any of the original or newly created variables, so no rows needed to be removed or filled in. This allowed us to keep the full dataset for analysis.

Next, the release_date column was converted into a datetime format so that we could extract the release year. A new variable called release_year was created to include release timing as a numeric feature in the models. Since the histograms showed that vote_count and popularity were strongly right-skewed, we also created log-transformed versions of these variables called log_vote_count and log_popularity. These transformations helped us better understand the distribution of the data during exploratory analysis.

The target variable for modeling was high_success. In the code, a title was classified as high success if its popularity was greater than or equal to the median popularity value in the dataset. Titles below the median were classified as low success. This created a nearly balanced target variable, with 41 high-success titles and 40 low-success titles. Because the classes were already balanced, no oversampling or undersampling was needed.

For the predictive models, we used vote_average, vote_count, release_year, on_netflix, on_hulu, and on_prime as the input features. Popularity was not used as a predictor because it was used to define the target variable. The data was then split into training and testing sets using a 75% training and 25% testing split, with stratification used to preserve the balance between high-success and low-success titles.

Methodology:

First we performed some statistical tests to learn more about the data

```
Kruskal-Wallis test for popularity across platforms
Statistic: 1.2703641693188672
p-value: 0.529839003979082

One-way ANOVA for popularity across platforms
Statistic: 0.5393858891905585
p-value: 0.5875044064479438

Kruskal-Wallis for vote_average
Statistic: 4.561282432177114
p-value: 0.10221864145958111

Kruskal-Wallis for vote_count
Statistic: 1.2135290184070584
p-value: 0.545111722887688
```

From these tests, we were able to learn that there were no statistically significant differences in popularity, vote average, or vote count across Netflix, Hulu, and Prime at the 5% significance level. The Kruskal-Wallis test for popularity had a p-value of about 0.530, and the one-way ANOVA test for popularity had a p-value of about 0.588. Since both values are greater than 0.05, we do not have enough evidence to say that average popularity differs significantly across the platforms.

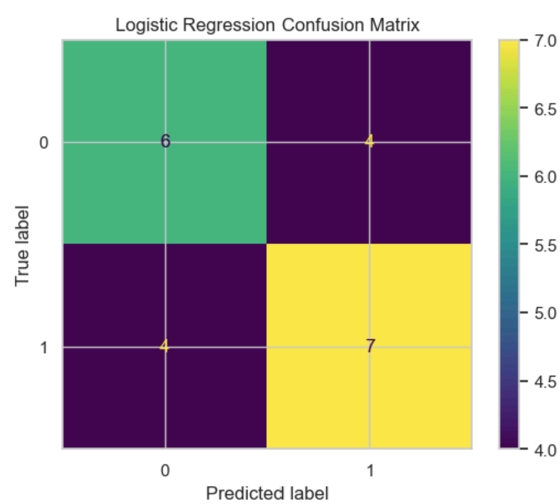
The Kruskal-Wallis test for vote average also had a p-value greater than 0.05, meaning there was no statistically significant difference in ratings across the three platforms. The same was true for vote count, where the p-value was also greater than 0.05. Although the boxplots and EDA graphs showed some visual differences between platforms, the statistical tests suggest that these differences may not be strong enough to be considered statistically significant in this dataset.

This is likely because the dataset is relatively small and the platform groups are not evenly represented. Therefore, the statistical tests should be interpreted with caution. Even though the tests did not show significant differences, the descriptive patterns and predictive models still provide useful insight into which platforms appear stronger based on title-level performance.

Logistic Regression

Logistic regression was used as the baseline classification model because it is simple, interpretable, and useful for predicting a binary outcome such as high success versus low success. The model was trained using the selected features: vote average, vote count, release year, and the three platform indicators.

```
=== Logistic Regression ===
Accuracy: 0.6190476190476191
F1: 0.6363636363636364
ROC-AUC: 0.6181818181818183
              precision    recall  f1-score   support

           0       0.60      0.60      0.60        10
           1       0.64      0.64      0.64        11

    accuracy                           0.62        21
   macro avg       0.62      0.62      0.62        21
weighted avg       0.62      0.62      0.62        21
```

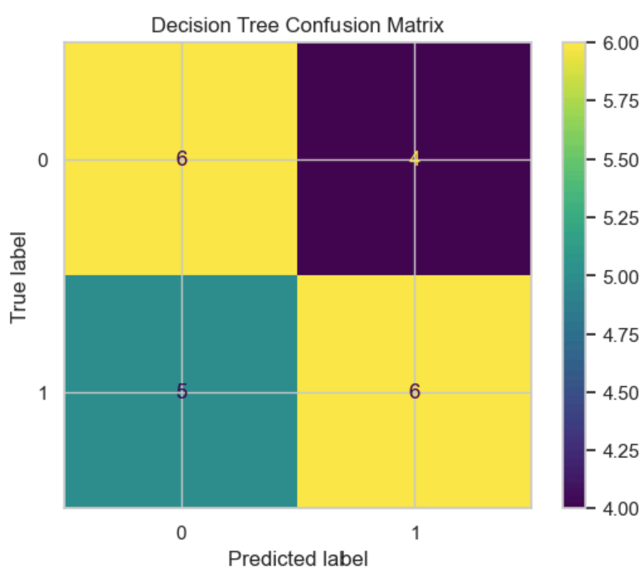


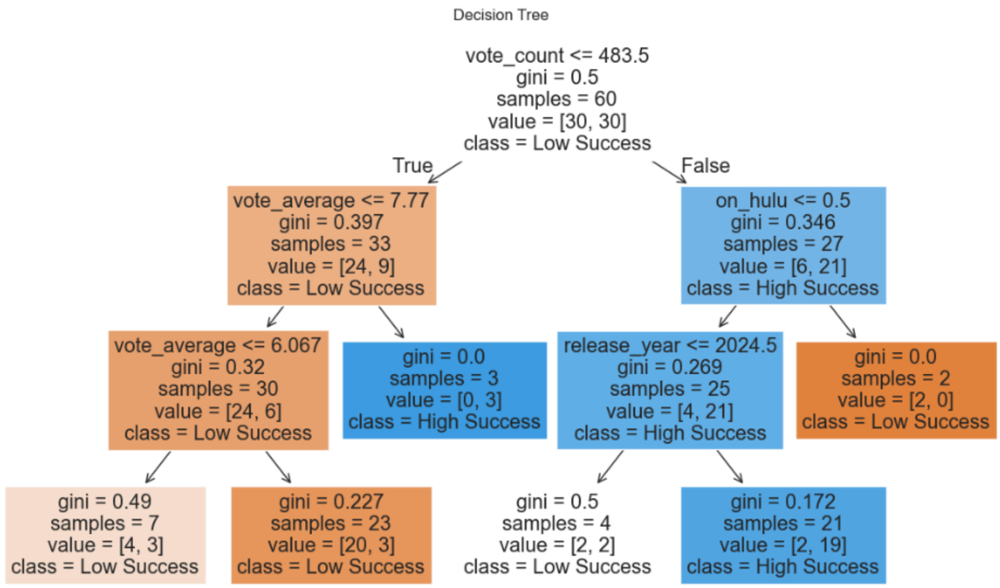
The logistic regression model achieved an accuracy of about 61.9%, an F1-score of about 0.636, and an ROC-AUC of about 0.618. This was the strongest test-set ROC-AUC among the models. The confusion matrix showed that the model correctly classified 6 out of 10 low-success titles and 7 out of 11 high-success titles. Overall, logistic regression performed the best on the test set and provided a useful baseline for comparing the other models.

Decision Tree

The decision tree model was used because it provides an interpretable rule-based approach to classification. Unlike logistic regression, a decision tree can capture nonlinear relationships and interactions between variables. The model was limited to a maximum depth of 3 to help reduce overfitting and keep the tree easier to interpret.

```
=== Decision Tree ===
Accuracy: 0.5714285714285714
F1: 0.5714285714285714
ROC-AUC: 0.5681818181818181
              precision    recall  f1-score   support

           0       0.55      0.60      0.57        10
           1       0.60      0.55      0.57        11

    accuracy                           0.57        21
   macro avg       0.57      0.57      0.57        21
weighted avg       0.57      0.57      0.57        21
```



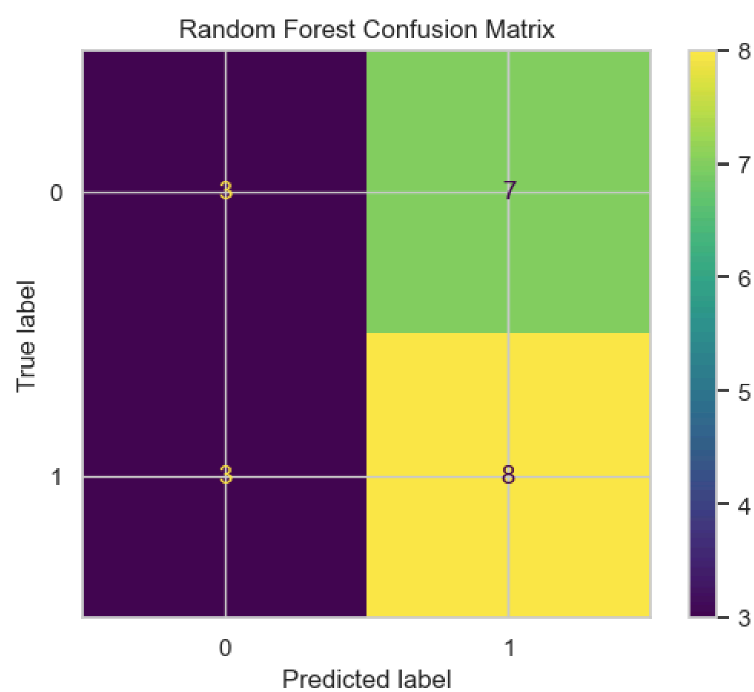


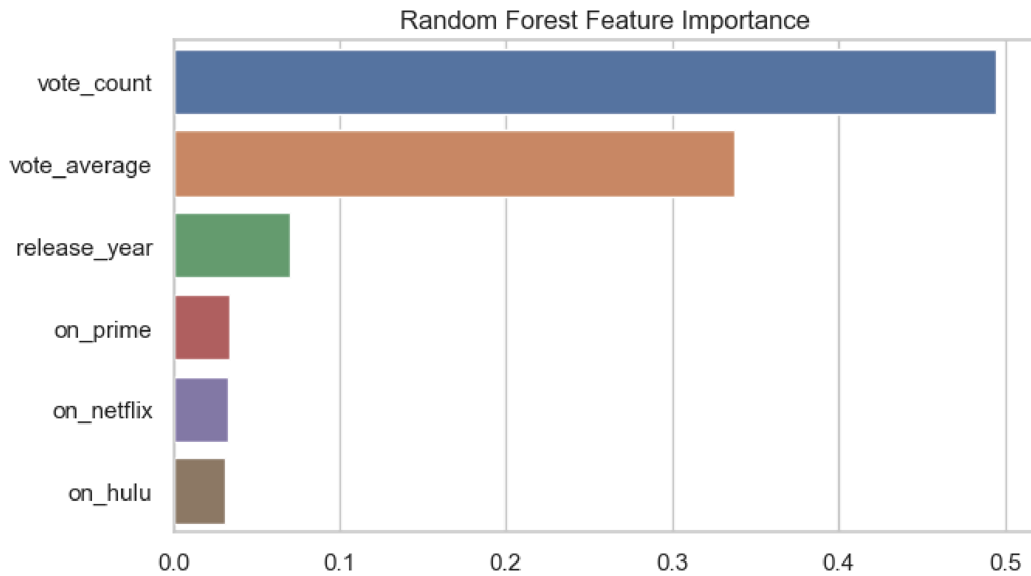
The decision tree achieved an accuracy of about 57.1%, an F1-score of about 0.571, and an ROC-AUC of about 0.568. This was slightly weaker than logistic regression, but the tree visualization was useful because it showed how the model split titles based on variables such as vote count, vote average, release year, and platform indicators. Overall, the decision tree was helpful for interpretation, but it was not the strongest model in terms of predictive performance.

Random Forest

The random forest model was used as a more advanced tree-based method. Unlike a single decision tree, random forest combines many trees to make predictions, which can improve stability and reduce overfitting. In this project, the random forest model used 200 trees with a maximum depth of 5.

```
=== Random Forest ===
Accuracy: 0.5238095238095238
F1: 0.6153846153846154
ROC-AUC: 0.6090909090909091
              precision    recall  f1-score   support

           0       0.50      0.30      0.38        10
           1       0.53      0.73      0.62        11

    accuracy                           0.52        21
   macro avg       0.52      0.51      0.50        21
weighted avg       0.52      0.52      0.50        21
```



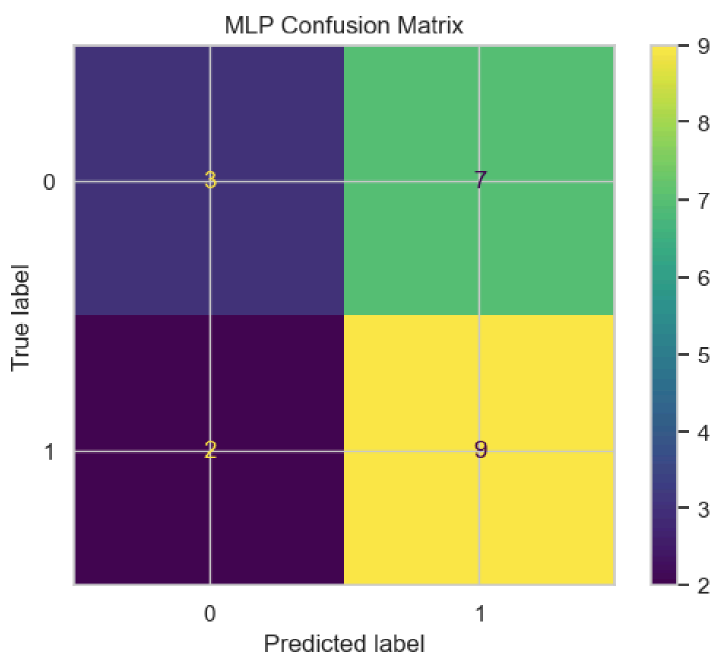


The random forest model achieved an accuracy of about 52.4%, an F1-score of about 0.615, and an ROC-AUC of about 0.609 on the test set. Although its accuracy was lower than logistic regression, its ROC-AUC was close to the logistic regression model. The random forest feature importance plot showed that vote_count was the most important predictor, followed by vote_average and release_year. The platform indicators had smaller importance values, which suggests that title-level performance measures were more important than platform membership alone.

<u>MLP / Feedforward Neural Network(Jordan Allwood)</u>

A multilayer perceptron (MLP), or feedforward neural network, was selected as an additional test because it is well suited for structured tabular data with a fixed set of input features, such as popularity, vote average, vote count, release year, and platform indicators. Unlike CNNs, which are designed for image data, or RNNs, which are intended for sequential or time-dependent data, an MLP is appropriate for this project because each movie is represented by a single row of numeric features rather than images or sequences. In addition, the MLP allows the analysis to include a neural-network-based method capable of capturing nonlinear relationships that simpler models such as logistic regression may miss. This makes it a useful comparison model for evaluating whether a more flexible deep learning approach can improve prediction of title success in the streaming dataset.

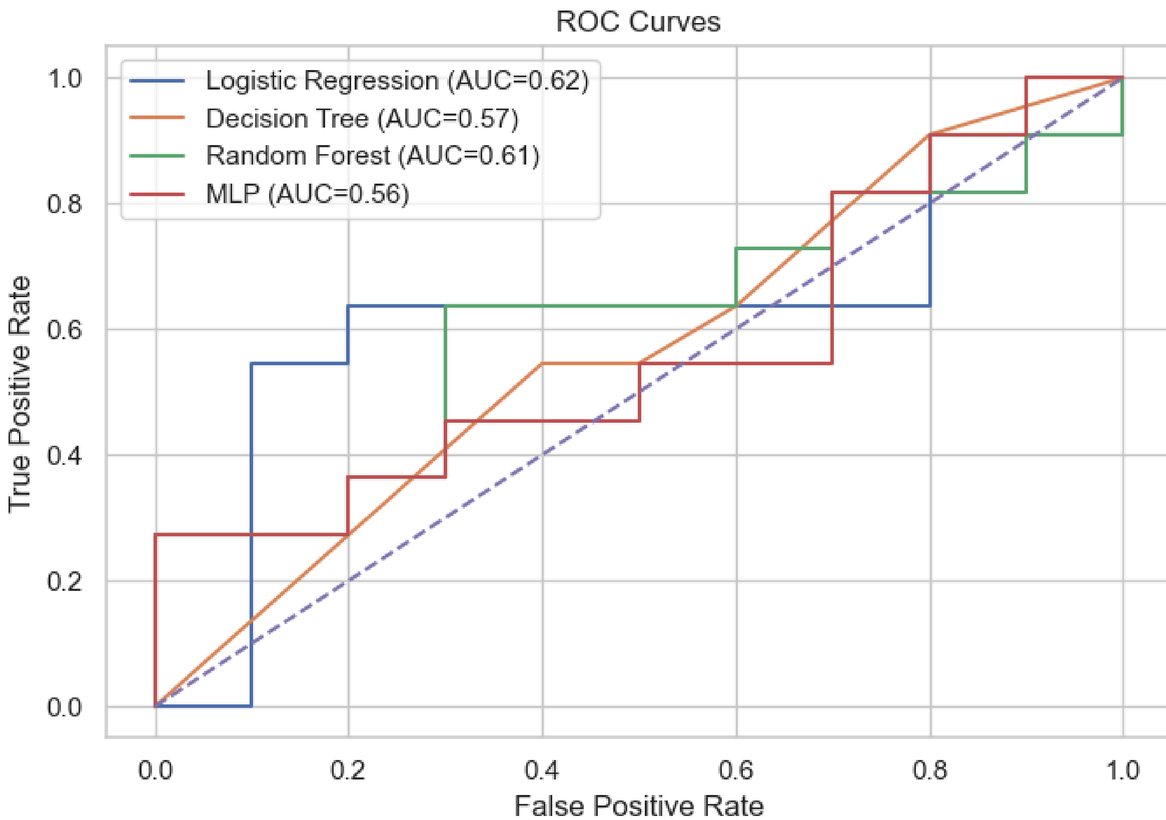
```
=== MLP / Feedforward Neural Network ===
Accuracy: 0.5714285714285714
F1: 0.6666666666666666
ROC-AUC: 0.5636363636363636
              precision    recall  f1-score   support

           0       0.60      0.30      0.40        10
           1       0.56      0.82      0.67        11

    accuracy                           0.57        21
   macro avg       0.58      0.56      0.53        21
weighted avg       0.58      0.57      0.54        21
```



The MLP, or feedforward neural network, showed moderate predictive performance in identifying whether a title would be classified as high success or low success. The model achieved an accuracy of 57.1%, an F1-score of 0.67 for the high-success class, and an ROC-AUC of 0.56, indicating that it performed only slightly better than random guessing overall. The confusion matrix shows that the model correctly identified 9 of 11 high-success titles, but only 3 of 10 low-success titles, meaning it was much better at detecting titles likely to be successful than those likely to be unsuccessful. This is also reflected in the class-specific recall values, where recall for high-success titles was 0.82, compared with only 0.30 for low-success titles.

In terms of the project goal, these results suggest that the MLP was somewhat useful for identifying titles with strong success potential, but it was less reliable for giving a balanced overall classification. The model appears to lean toward predicting the high-success class, which may be helpful when the goal is to highlight titles or platforms with stronger upside potential, but it also means that weaker-performing titles were often misclassified as successful. Compared with the earlier models, the MLP did not provide the strongest overall discrimination, but it does reinforce the broader finding that there is at least some detectable pattern in the title-level features. For the investment question, this means the MLP supports the idea that certain platforms may contain more titles with strong success profiles, but its moderate and somewhat imbalanced performance suggests that it should be treated as a supporting model rather than the main basis for the final platform recommendation.

ROC Curves



The ROC curves were used to compare how well each model separated high-success titles from low-success titles. Based on the test-set ROC-AUC values, logistic regression performed the best with an AUC of about 0.62, followed by random forest at about 0.61, decision tree at about 0.57, and MLP at about 0.56.

5-Fold CV ROC-AUC

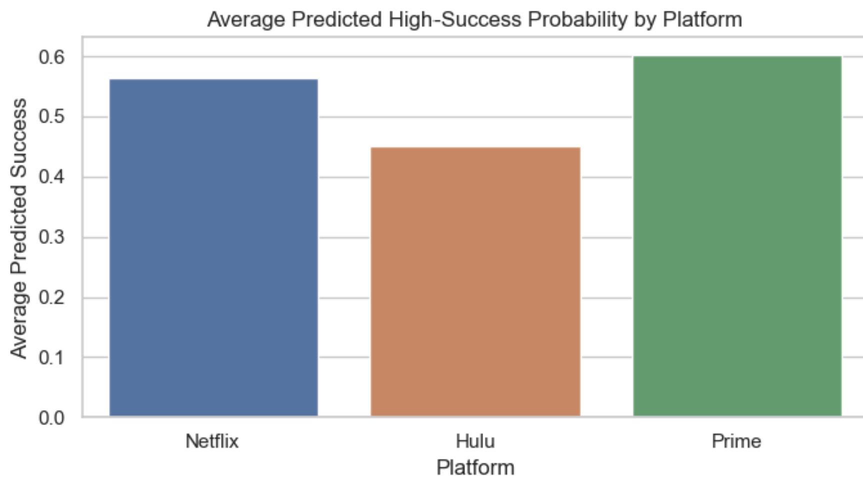
```
=== 5-Fold CV ROC-AUC ===
Logistic Regression: 0.679861111111111 +/- 0.14405580060612486
Decision Tree: 0.6602430555555555 +/- 0.07434448616714484
Random Forest: 0.6895833333333334 +/- 0.14981904594469214
MLP: 0.5690972222222223 +/- 0.1811135874489873
```

To evaluate the models more reliably, we also used 5-fold cross-validation with ROC-AUC as the scoring metric. In cross-validation, random forest had the highest average ROC-AUC at about 0.690, followed closely by logistic regression at about 0.680 and decision tree at about 0.660. The MLP had the lowest cross-validation ROC-AUC at about 0.561. This suggests that logistic regression performed best on the specific test split, while random forest performed slightly better across multiple folds.

Overall, the model results show moderate predictive ability. The models were able to detect some patterns in the data, but the prediction scores were not extremely high. This suggests that the available variables are useful, but they do not fully explain title success.

Platform Results

```
Platform Summary:
  Platform  Average Predicted Success  Average Popularity  \
2    Prime                   0.601333           50.421273
0  Netflix                   0.564666           32.985571
1     Hulu                   0.449840           30.716744

   Average Vote Average
2              6.412818
0              7.044429
1              6.995778
```



After building the models, we used the random forest model to estimate the average predicted probability of high success for each platform. Prime had the highest average predicted success probability at about 0.601, followed by Netflix at about 0.565, and Hulu at about 0.450.

This suggests that Prime had the strongest predicted success signal in this dataset. Netflix was close behind and also showed strong overall performance, especially in average vote rating. Hulu had the lowest predicted success probability, which suggests that it appeared weaker than Prime and Netflix based on the available title-level features.

The platform results support the overall finding from the EDA that Prime and Netflix appear stronger than Hulu in this sample. However, these results should be interpreted carefully because the dataset is small and does not include all possible measures of streaming platform success.

Conclusion/Future Work:

In conclusion, this project showed that predictive modeling can be a useful way to evaluate streaming platform strength based on title-level performance. Across the exploratory analysis and model results, Prime showed the strongest content-based success signal, with Netflix close behind and Hulu ranking lower in this dataset. Prime stood out with the highest average predicted high-success probability and strong popularity patterns, while Netflix showed strong audience ratings and consistent overall performance. These results suggest that both Prime and Netflix are well positioned in terms of the titles represented in this dataset.

The modeling results also showed that title-level engagement and rating variables were important for predicting success. Vote count, vote average, and release year contributed more to the models than platform membership alone. This suggests that a platform's long-term strength is closely connected to the quality and audience engagement of the content it offers. Logistic regression performed best on the test set, while random forest performed strongest during 5-fold cross-validation and provided helpful feature importance results. The decision tree gave a clear visual interpretation of the prediction process, and the MLP added a deep-learning comparison to the analysis.

For future work, this project could be expanded by using a larger dataset with more titles, more release years, and additional streaming platforms. Adding features such as genre, runtime, content type, original versus licensed content, subscriber trends, and platform revenue could create a more complete picture of streaming platform performance. Future versions of the project could also compare movies and TV shows separately, since success may look different across different types of content. More advanced models such as gradient boosting, XGBoost, or larger neural networks could also be tested with a bigger dataset.

Overall, this project successfully identified useful patterns in streaming title performance and connected those patterns to platform-level success. The results support the idea that data science methods can help compare streaming platforms beyond simple popularity rankings by combining audience engagement, ratings, release timing, and predictive modeling.

References:

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https://www.kaggle.com/datasets/razvanmihaihanghicel/streaming-platform-content-availability-2026

Distribution of Work:

Introduction/Background: Jordan

EDA: Jordan

Data processing: Meghana

Methodology: Meghana/Jordan

Conclusion/Future Work: Meghana